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Functional connectivity in whole-brain and network analysis differentiates minimally conscious from unresponsive patients: a resting-state fNIRS study

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Abstract

Background Accurately assessing and detecting residual awareness in patients with vegetative state/unresponsive wakefulness syndrome (VS/UWS) and minimally conscious state (MCS) remains significant challenges. We aimed to investigate the whole-brain and network characteristics based on resting-state functional near-infrared spectroscopy (fNIRS) in patients with disorders of consciousness (DOC). Additionally, we sought to identify specific biomarkers to differentiate MCS from VS/UWS and evaluate their classification performance.

Methods For DOC patients, the Coma Recovery Scale-Revised (CRS-R) assessment was conducted, and all participants underwent a 5-min resting-state fNIRS recording. Functional connectivity features based on region of interest, channel, and network analyses were calculated to explore significant differences between healthy adults and DOC, as well as MCS patients and VS/UWS patients. Pearson correlation analysis was then performed to examine the relationship between fNIRS features and CRS-R scores. Receiver operating characteristic analysis and linear support vector machines were employed to assess classification performance.

Results We included 52 DOC patients ($n = 26$ for MCS and $n = 26$ for VS/UWS) and 49 healthy controls in the final analysis. Compared to healthy controls, DOC patients showed widespread impairments in both whole-brain and network-based functional connectivity. Additionally, VS/UWS patients exhibited significantly reduced functional connectivity compared to MCS patients, including connectivity between the prefrontal cortex, premotor cortex, sensorimotor regions, and Wernicke's area ($p < 0.01$), as well as within auditory, frontoparietal, and default mode network ($p < 0.05$). Some of these connectivity differences were found to correlate with the total CRS-R score, as well as the visual, motor, and verbal subscale scores ($p < 0.05$). In terms of classification performance for distinguishing MCS

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from VS/UWS patients, the functional connectivity between channel 4 and channel 29 showed the highest accuracy among whole-brain features, with a classification accuracy of 76.92% and an area under the curve (AUC) of 0.818. Among the resting-state network features, the auditory network exhibited the highest accuracy, achieving 73.08% with an AUC of 0.803.

Conclusion fNIRS could effectively detect abnormal brain network functional connectivity in DOC patients and could provide valuable insights for differentiating MCS and VS/UWS patients.

Keywords Disorders of consciousness, Resting-state fNIRS, Functional connectivity, Brain network, Minimally conscious state, Unresponsive wakefulness syndrome

Background

Disorders of consciousness (DOC) are characterized by impairments in wakefulness and awareness [1], typically resulting from severe acquired brain injuries. Prolonged DOC is defined as unconsciousness lasting longer than 28 days [1]. DOC is primarily categorized into coma, vegetative state (VS)/unresponsive wakefulness syndrome (UWS) and minimally conscious state (MCS) [2]. VS/UWS patients are clinically awake but lack awareness of themselves and their surrounding environment, while MCS patients exhibit minimal, fluctuating, but reproducible purposeful evidence of awareness [3]. Distinguishing between VS/UWS and MCS is crucial for guiding treatment and prognostic decisions [4], as MCS patients exhibit signs of consciousness and may show a greater potential for awareness [5]. Currently, bedside behavioral assessment remains the clinical reference standard for evaluating awareness in DOC patients [6]. However, it has been reported that approximately 41% of patients were misdiagnosed based on behavioral observations [7]. Accurately evaluating brain function and residual awareness remains a complex challenge due to fluctuations in vigilance, cognitive and motor deficits, as well as examiner and environmental factors [8].

Functional neuroimaging, as an objective technique, can help bridge this gap by providing essential information about the state of consciousness. A variety of studies have provided valuable insights into evaluating awareness and exploring structural and functional characteristics through positron emission tomography (PET), functional magnetic resonance imaging (fMRI), and electroencephalography (EEG) in both resting and task-related states [4, 9]. The task-state paradigms, which involve sensory stimulation and active commands, have not been standardized. Due to potential motor and language deficits, the likelihood of false-negative results increases. Moreover, these paradigms are often limited by the technical resources, whereas resting-state assessments require no complex setups or active participation from the subject [10].

A variety of studies have conducted resting-state analysis to assess patients with DOC. Demertzi et al. [11] demonstrated that connectivity values within the

auditory network could reliably distinguish 20 out of 22 patients using fMRI. Additionally, Stefan et al. [12] used high-density EEG and complex network analysis to index consciousness and predict outcomes. However, fMRI is expensive, immobile, susceptible to motion artifacts, which limits its applicability in critically ill populations [13]. While EEG is portable, its low spatial resolution and susceptibility to environmental noise make it challenging in bedside settings. These drawbacks underscore the need for the developing alternative neuroimaging techniques for evaluating consciousness.

Functional near-infrared spectroscopy (fNIRS) is a portable, cost-effective neuroimaging technique with good temporal and spatial resolution [14]. In contrast to fMRI, fNIRS is more robust to motion artifacts and the presence of metal implants [8]. It estimates brain activity through neurovascular coupling by estimating concentration changes in oxygenated (HbO) and deoxygenated (HbR) hemoglobin [8]. While fNIRS has been widely used in cognitive function assessment, its application in evaluating prolonged DOC remains underexplored [15]. Existing resting-state studies based on fNIRS are limited and fragmented, with small sample sizes. Some studies [13] have detected resting-state networks, such as the sensorimotor, auditory, frontoparietal, and DMN networks, while others [16] have highlighted impaired connectivity in VS/UWS and MCS patients. However, previous fNIRS studies have demonstrated differences between DOC patients and healthy controls, but lacked the fNIRS differential parameters between MCS and VS/UWS. Additionally, current studies have focused on the frontal region, and there is limited information available on whole-brain or network-based fNIRS characteristics in DOC patients. In summary, while fNIRS shows potential for advancing the diagnosis and prognosis of DOC, further research is required.

Therefore, we conducted a resting-state fNIRS study involving DOC patients and healthy controls (HC) with the following objectives: (1) To investigate the differences in whole-brain fNIRS characteristics between HC and DOC patients, as well as between MCS and VS/UWS patients, and explore the underlying mechanisms of DOC; (2) To assess the effectiveness of fNIRS in

identifying functional connectivity within brain networks and the contribution of different brain networks to consciousness; (3) To identify specific fNIRS biomarkers that differentiate MCS from VS/UWS and evaluate classification accuracy.

Methods

Participants and study design

This cross-sectional study included DOC patients and healthy controls. DOC patients were recruited from the Department of Rehabilitation at The Third Affiliated Hospital of Sun Yat-Sen University, and HC were recruited from the community between December 2023 and January 2025. Inclusion criteria for patients were as follows: disorders of consciousness lasting more than 28 days, resulting from stroke, traumatic brain injury or hypoxic-ischemic encephalopathy, with stable vital signs. All patients were diagnosed as either VS/UWS or MCS based on repeated CRS-R assessments (≥ 5), conducted by two professional therapists who were blinded to the fNIRS results, within a 2 week period [17]. One CRS-R assessment was performed on the same day as the resting-state fNIRS acquisition. Exclusion criteria included: (1) Age < 18 years; (2) Intracranial hypertension, severe hydrocephalus, intracranial infection, and skull defect; (3) Other serious medical conditions that could interfere with the study. Ethical approval for this study was granted by the Research Ethics Board of The Third Affiliated Hospital of Sun Yat-Sen University. Written informed consent was obtained from all healthy participants, as well as from the legal guardians or family members of the DOC patients.

The CRS-R scale consists of six functional subscales: auditory, visual, motor, oromotor/verbal, communication, and arousal, with a total of 23 items organized hierarchically [17, 18]. In addition to the total score, the individual subscale scores are crucial for diagnosing the state of consciousness. Furthermore, patient-related information, including gender, age, etiology, affected hemisphere, lesion location, and time since injury, was collected as covariates.

Data acquisition and preprocessing

In this study, the NirSmart-6000A system (Danyang Huichuang Medical Equipment Co., Ltd., China), based on the continuous wave principle, was used to continuously monitor and record brain HbO and HbR levels during the task. The system consists of a near-infrared light source and avalanche photodiode detectors operating at wavelengths of 730 nm and 850 nm, with a sampling rate of 11 Hz. The setup includes 24 light sources and 24 detectors, forming 63 effective channels, with an average source-detector distance of 3 cm (ranging from 2.7 to 3.3 cm), regarding the international 10/20 system

for positioning. Based on the Brodmann Areas defined in Chris Rorden's MRICro tool [19], the channels were assigned to five regions of interest (ROIs): the prefrontal cortex (FPC), pre-motor cortex (PMC), Wernicke's area, sensorimotor cortex (SMC), and visual cortex (VC). Channel 23 was excluded as it did not correspond to either hemisphere of the brain (Fig. 1). Detailed information for all channels is provided in Additional file 1: Table S1. The lowest row of probes was aligned with the subject's eyebrow arch, while the middle row was parallel to the midsagittal line from the nasal root to the occipital tuberosity [20]. A 5-min resting-state fNIRS scan was performed in a quiet room without external stimuli. The patient's bed was elevated to a 30° angle, with a pillow placed behind the back and neck to support the head, ensuring the optodes covered the entire head. Arousal stimulation was applied to the subjects before the assessment and ensured they remained still during the evaluation.

All fNIRS data preprocessing, analysis and visualization were performed using MATLAB (R2022a, The MathWorks Inc., USA) [21] with the Homer2 Toolbox [22], NIRS-KIT [23], and BrainNet Viewer [24]. The preprocessing was as follows: (1) Trimming the first 10 s; (2) Calculating Coefficient of Variation (CV), which was commonly used in multi-channel fNIRS measurements [8, 25]. Specifically, CV estimated the signal-to-noise ratio by calculating the ratio of the standard deviation to the mean of the raw signal data (intensity changes). The formula was as follows:

$$CV = \frac{\sigma}{\mu} \cdot 100\%$$

Here σ was the standard deviation of the channel data, and μ was the mean of the channel. A higher CV typically indicated poor signal quality, which might be due to motion artifacts or blood flow-related changes. For each channel, a CV value > 20% was considered a bad channel. If the number of bad channels was $\geq 30\%$ for a participant, the participant's data were deemed unacceptable and excluded from further analysis; (3) Converting raw data to the optical density (OD) [26]; (4) Correcting motion artifacts by spline interpolation algorithm [27]; (5) 0.01–0.1 Hz bandpass filtering to minimize the physiological interference and achieve optimal signal-to-noise ratio, targeting blood pressure (Mayer) waves (~ 0.1 Hz), respiration (~ 0.4 Hz) and heart pulsation (1–1.5 Hz) [28]; and (6) Converting optical density to relative concentrations of HbO/HbR using the modified Beer-Lambert law, with differential path length factor of 6.0. Total hemoglobin (HbT) was computed as the sum of HbO and HbR.

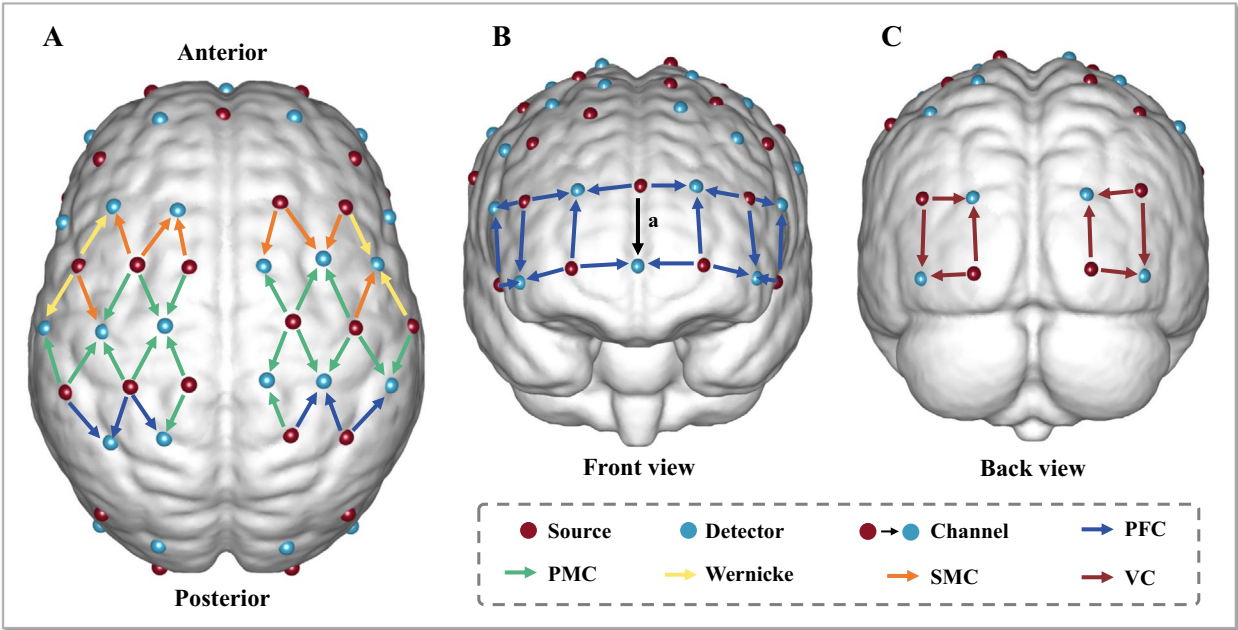


Fig. 1 Configuration of fNIRS optodes. **A** Top view; **B** Front view; **C** Back view. The definition of ROI was based on Brodmann Area (Chris Rorden's MRIcro). Each ROI was divided into left and right cortex for statistical analysis. ^a Channel 23 was removed from the ROI analysis as it did not belong to either side of the brain. Abbreviation: ROI, region of interest; PFC, prefrontal cortex; PMC, pre-motor cortex; SMC, sensorimotor cortex; VC, visual cortex

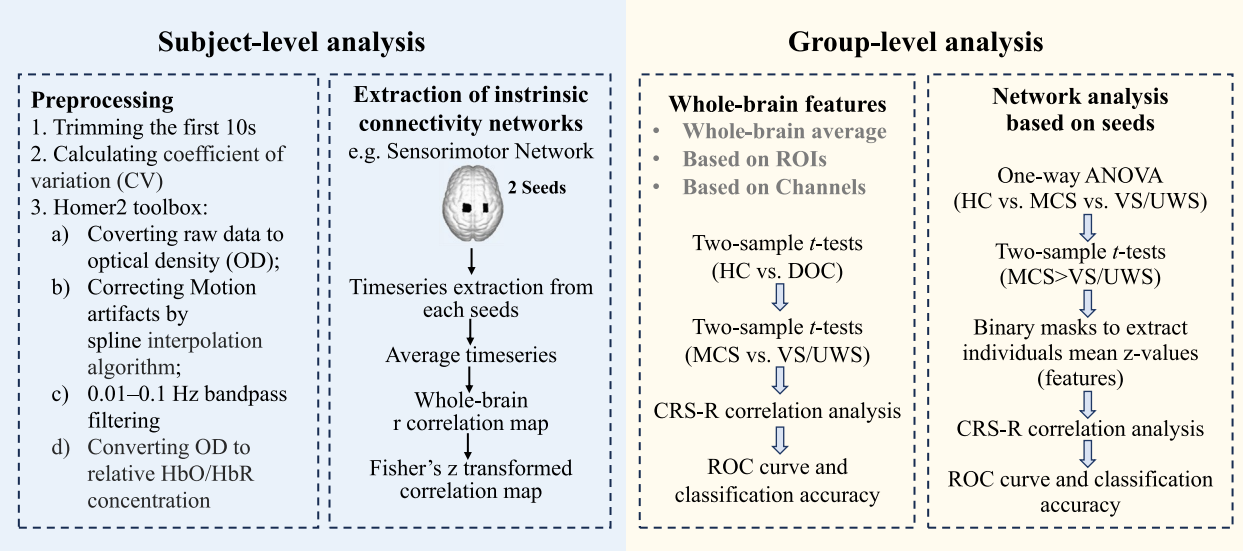


Fig. 2 Analysis pipeline. Abbreviation: CV, Coefficient of Variation; OD, optical density; HbO, oxygenated hemoglobin; HbR, deoxygenated hemoglobin; ROI, region of interest; HC, healthy controls; DOC, disorder of consciousness; VS, vegetative state; UWS, unresponsive wakefulness syndrome; MCS, minimally conscious state; CRS-R, Coma Recovery Scale-Revised; ROC, Receiver Operating Characteristic; ANOVA, Analysis of Variance

Only HbT was used to extract functional connectivity networks since previous work has shown that HbT is more robust than either HbO or HbR [29].

Whole-brain connectivity analysis

Data analysis is illustrated in Fig. 2. The correlation coefficients (r) between each pair of channels were computed using Pearson correlation analysis, resulting in a

functional connectivity (FC) matrix. Fisher's z-transformation was then applied to convert the correlation coefficients into z-values for normalization.

Whole-brain features consisted of the following three features: (1) whole-brain average FC; (2) ROI-based FC, and (3) Channel-based FC. The z-value matrix for each subject was averaged to obtain the whole-brain average FC. For the ROI-based FC analysis, blood oxygenation

level-dependent time series were extracted from each ROI, and the time series for all channels within a given ROI were averaged. The correlation coefficients between the time series of every two ROIs were calculated, resulting in an ROI-ROI FC matrix for each individual. For channel-based FC, correlation coefficients between the time series of every two channels (Ch) were calculated, producing a 63×63 Ch-Ch FC matrix for each individual.

Network connectivity analysis

Five networks were selected for analysis: sensorimotor, auditory, frontoparietal, default mode, and visual. These networks contain cortical nodes, making them well-suited for fNIRS evaluation [13]. The sensorimotor, auditory, and visual networks are associated to lower-level processing, whereas the frontoparietal and default mode networks are involved in higher-level functions like executive processing and awareness [11, 30]. Moreover, these networks have been extensively researched in patients with both acute and chronic disorders of consciousness [31], providing valuable diagnostic and prognostic value [11, 32].

A well-established seed-based correlation method was employed to evaluate network FC [33, 34]. Multiple channels were selected as seeds for each network to reduce the impact of abnormal signals from brain injury or poor optode-scalp coupling (Additional file 1: Fig. S1) [35]. Additionally, while the chosen seed might not always represent the entire network, using multiple seed regions

helped address this issue and ensured accurate network characterization in patients [11]. Seed locations were selected based on previous fMRI studies and are listed in Table 1 and Additional file 1: Fig. S1 [11, 13]. Seed-correlation analysis involved extracting HbT time series from each seed, averaging the HbT times series for each network, and using these to estimate whole-brain correlation maps. These maps were then transformed using Fisher’s z transformation for group-level comparisons.

Statistical analysis

Statistical analyses were performed using MATLAB R2022a, The MathWorks Inc., USA) [21] and SPSS (version 26.0, IBM Corp., Armonk, NY, USA) [36]. Clinical characteristics and variables are presented as means $\pm$ standard deviations, medians (interquartile ranges), or counts (percentages). All results were considered significant $p < 0.05$, corrected for multiple comparisons using the false discovery rate (FDR). The general statistical analysis workflow is outlined in Fig. 2.

To investigate the first aim of the study, i.e. to investigate the differences in whole-brain fNIRS characteristics between HC and DOC patients, as well as between MCS and VS/UWS patients, a whole-brain analysis was performed. First, one-way ANOVA was conducted to compare the whole-brain average z-values among the HC, MCS and VS/UWS groups. Next, two-sample t tests were conducted to compare the ROI-based FC, and channel-based FC between the HC and DOC groups,

Table 1 Positions of seeds and their associated connections in resting-state seed-based network connectivity analysis for healthy participants and DOC patients

Network name	Seed location (Channel)	Connected brain regions ^a	Z-value ^b		
			HC	MCS	VS/UWS
Sensorimotor network	Right precentral gyrus (32)	Left and right pre and post-central gyrus, Parietal cortices, Frontal cortices	0.85 $\pm$ 0.36 ^{**}	0.56 $\pm$ 0.35	0.36 $\pm$ 0.30
	Left precentral gyrus (54)				
Auditory network	Right supramarginal gyrus (2)	Left and right supramarginal gyrus, Left superior and middle frontal cortices, Left and right pre- and postcentral gyrus	0.82 $\pm$ 0.31 ^{**}	0.63 $\pm$ 0.39 [#]	0.38 $\pm$ 0.25
	Left supramarginal gyrus (13)				
Frontoparietal network	Right middle frontal gyrus (20)	Left and right frontal cortices, Precen-tral cortices, Parietal cortices	0.87 $\pm$ 0.29 ^{**}	0.68 $\pm$ 0.36 [#]	0.42 $\pm$ 0.27
	Left inferior frontal triangularis (12)				
	Right inferior parietal gyrus (44)				
	Left inferior parietal gyrus (15)				
Default mode network	Right superior orbital frontal gyrus (6)	Left and right medial and superior pre-frontal cortices, superior and inferior parietal cortices	0.98 $\pm$ 0.24 ^{**}	0.7 $\pm$ 0.35 [#]	0.45 $\pm$ 0.29
	Left superior frontal gyrus (8)				
	Right superior parietal gyrus (46)				
	Left superior parietal gyrus (39)				
Visual network	Right superior occipital gyrus (60)	Left and right middle and superior oc-cipital gyrus, superior parietal cortices, Postcentral gyrus	0.80 $\pm$ 0.28 ^{**}	0.51 $\pm$ 0.39	0.35 $\pm$ 0.27
	Left middle occipital gyrus (57)				

^aRegions connected at a sparsity threshold of 40% for the five networks in healthy participants
^bThe z-values of the seeds to whole-brain channels for each network were averaged across all channels for each subject, and a one-way ANOVA was then conducted to compare the three groups
DOC, disorders of consciousness; HC, healthy controls; MCS, minimally conscious state; VS, vegetative state; UWS, unresponsive wakefulness syndrome
^{*} $p < 0.05$ compared with MCS group, after Bonferroni correction; [#] $p < 0.05$ compared with UWS group, after Bonferroni correction

and the MCS and VS/UWS groups. In the HC and DOC group analysis, gender and age were considered as confounding factors, while in the MCS and VS/UWS group comparison, gender, age, etiology, and time since injury were included as covariates. Previous studies [37, 38] demonstrated that age and gender significantly influence neurovascular response and brain activation patterns. Etiology and time since injury were identified as key factors affecting the prognosis and brain network reorganization of DOC patients [39, 40]. Given the heterogeneity in DOC patients, incorporating these covariates helped enhance the robustness of the analysis and control for potential confounders. Correlation analysis was then performed between the significant features of the MCS and VS/UWS groups and the CRS-R scores, with Pearson correlation coefficients calculated. The receiver operating characteristic (ROC) curve [41] was generated to calculate the area under the curve (AUC) to quantify the performances of these features in distinguishing MCS from VS/UWS. To identify which whole-brain features most effectively differentiate between MCS and VS/UWS patients, a single-feature linear support vector machine (SVM) classifier [42] was employed. A 9×1 vector was generated for each subject (i.e., a 52×9 matrix), encompassing nine whole-brain features: the average z-value, three ROI-to-ROI features, and five Ch-Ch features. A leave-one-out cross-validation procedure [11, 13, 43] was used to evaluate the classifier, utilizing a set of 52 feature vectors extracted from patients with MCS ($n=26$) and VS/UWS ($n=26$). This method allowed us to assess the ability of each feature to predict the consciousness state of each patient, using the remaining patients as the training set. The classifier's performance was assessed by calculating the true positives (MCS), true negatives (VS/UWS), and overall classification accuracy.

The network analysis followed these steps, based on previous fMRI studies [11]: (1) The data from each network in the HC, MCS, and VS/UWS groups were averaged across all channels for each subject. A one-way analysis of variance was then performed to compare the three groups, using HC as the reference for accurate network characterization; (2) Two-sample t tests were conducted to identify regions with higher FC in the MCS group compared to the VS/UWS group for each network; (3) The resulting difference maps were saved as masks to extract mean z-values (average z-values across the mask) for each individual within each network, representing network features; (4) Two-sample t tests were conducted between MCS and VS/UWS groups for each network feature; (5) A correlation analysis was conducted between CRS-R scores and network features; (6) To evaluate the effectiveness of the network features in distinguishing MCS from VS/UWS, ROC curves, AUC, and an SVM classifier using leave-one-out cross-validation were

employed to assess the classification performance of the network features in differentiating MCS patients from VS/UWS patients.

Steps 1 to 5 were employed to address the second aim of the study, which was to assess the effectiveness of fNIRS in identifying functional connectivity networks and the contribution of different brain networks to consciousness. The generation of ROC curves and calculation of classification accuracy were used to achieve the third objective of the study, i.e. identifying specific fNIRS biomarkers for distinguishing MCS from VS/UWS.

Results

Clinical characteristics

Four DOC patients (7%) and one healthy control (2%) were excluded due to bad channels exceeding 30%. Ultimately, 52 DOC patients and 49 healthy controls were included in the final analysis. Among the DOC patients, 26 were diagnosed with VS/UWS (mean age: 59.7 ± 13.9 years, 11 females) and 26 with MCS (mean age: 59.1 ± 12.1 years, 8 females). The median (interquartile range, IQR) time since injury was 5.2 (6.5) months for VS/UWS patients and 4.2 (7.8) months for MCS patients. The primary etiology for these patients was cerebral hemorrhage, which accounted for 25 patients. Table 2 provides the detailed demographics and clinical features of each DOC patient.

Whole-brain analysis

The whole-brain features of the HC and DOC groups are presented in Additional file 1: Fig. S2. One-way ANOVA revealed a significance difference in whole-brain average z-values among HC group, MCS group and VS/UWS group ($F(2, 98) = 27.04$, $p < 0.001$, Additional file 1: Fig. S2A). Bonferroni-corrected post-hoc tests showed significant differences between the HC and MCS groups ($p < 0.001$) and between the HC and VS/UWS groups ($p < 0.001$). Additionally, patients with MCS exhibited significantly higher whole-brain z-values compared to those with VS/UWS ($p = 0.023$ after Bonferroni correction, Additional file 1: Fig. S2A). The DOC group showed notably lower FC based on both ROI and channel analyses (after FDR correction, Additional file 1: Fig. S2B, C, D), particularly in the PFC, PMC, SMC, and Wernicke's areas, indicating a global deficit in FC among patients with DOC.

The group differences in both ROI-based and channel-based FC between MCS and VS/UWS patients, without adjustment for any covariates, are illustrated in Additional file 1: Fig. S3. In contrast, Fig. 3 presents the group differences after controlling for potential confounding factors, including gender, age, etiology, and time since injury. For ROI-based FC, the significant connections identified before adjustment remained consistent after

Table 2 The characteristics of patients with disorders of consciousness

Number	Gender	Age	Etiology	MRI/CT findings	Time since injury (months)	Diagnosis	CRS-R scores ^a
1	Male	36	CH	Left hemisphere extensive lesions	44.6	MCS-	9 (1\2\2\2\0\2)
2	Male	83	TBI	Bilateral frontal lobes, left parietal, insular lobes lesions	9.5	UWS	6 (1\1\1\1\0\2)
3	Female	60	TBI	Bilateral frontal, temporal, left occipital lobe lesions	7.9	MCS-	9 (1\3\3\1\0\1)
4	Male	58	CI	Brainstem lesions	5.2	MCS-	9 (2\3\2\1\0\1)
5	Male	48	CH	Right thalamus, left frontal lobe, basal ganglia lesions	2.5	MCS-	15 (2\4\5\2\0\2)
6	Female	67	CI	Bilateral cerebral hemispheres extensive lesions	4.5	UWS	11 (1\1\5\2\0\2)
7	Female	60	CH	Right frontal, parietal lobes lesions	1.7	MCS-	8 (1\1\3\1\0\2)
8	Female	71	CH	Left basal ganglia lesions	9.2	MCS-	10 (0\1\5\2\0\2)
9	Male	42	CH	Left basal ganglia lesions	1.1	MCS-	11 (2\1\5\1\0\2)
10	Female	73	CH	Left basal ganglia lesions	1.3	UWS	8 (2\1\2\1\0\2)
11	Male	65	CH	Right frontal, temporal, insular lobes, left thalamus, basal ganglia lesions	3.8	MCS-	13 (2\3\5\1\0\2)
12	Female	63	CH	Left parietal lobe lesions	1.1	MCS-	9 (0\1\5\1\0\2)
13	Male	71	CH	Bilateral frontal, parietal, temporal lobes lesions	1.0	MCS-	10 (1\1\5\1\0\2)
14	Male	76	CI	Bilateral cerebral hemispheres lesions	1.3	MCS-	11 (2\1\5\1\0\2)
15	Male	58	CH	Left basal ganglia lesions	5.7	MCS-	11 (1\1\5\2\0\2)
16	Male	42	CH	Ventricular hemorrhage with subarachnoid hemorrhage	1.8	MCS-	10 (1\1\5\2\0\1)
17	Male	71	CI	Bilateral cerebral hemispheres extensive lesions, cerebellum, pons lesions	5.7	MCS-	6 (1\0\3\1\0\1)
18	Male	59	TBI	Right hemisphere extensive lesions, brainstem lesions	5.6	UWS	7 (1\1\2\1\0\2)
19	Male	78	CH	Bilateral frontal, parietal lobes, basal ganglia lesions	10.1	MCS-	9 (1\3\2\1\0\2)
20	Female	48	HIE	Bilateral cerebral cortex extensive lesions, brainstem lesions	2.5	MCS-	10 (1\0\5\2\0\2)
21	Male	54	CH	Left frontal, temporal, parietal lobes, basal ganglia lesions	11.0	UWS	8 (1\1\2\2\0\2)
22	Male	75	CH	Widespread subarachnoid hemorrhage	1.2	UWS	7 (1\1\2\1\0\2)
23	Male	59	CH	Left temporal, parietal lobes, basal ganglia lesions	1.2	MCS-	8 (2\1\2\1\0\2)
24	Male	63	CH	Left thalamus lesions	2.6	MCS-	9 (1\3\2\1\0\2)
25	Male	48	CH	Pontine lesions	7.5	MCS-	9 (1\3\2\1\0\2)
26	Female	77	CI	Bilateral cerebral hemispheres extensive lesions	1.3	MCS-	10 (1\1\5\1\0\2)
27	Male	38	CH	Left pons, right thalamus lesions	3.1	MCS+	11 (2\3\2\2\1\1)
28	Male	52	TBI	Bilateral frontal, left temporal lobes lesions	9.8	MCS+	12 (3\3\3\2\0\1)
29	Female	54	CI	Bilateral cerebral cortex extensive lesions	6.3	UWS	6 (1\1\2\0\0\2)
30	Male	74	TBI	Bilateral cerebral hemispheres lesions	35.4	UWS	4 (0\0\2\1\0\1)
31	Male	68	TBI	Bilateral frontal, temporal, parietal lobes lesions	5.3	UWS	4 (0\0\2\1\0\1)
32	Female	56	CH	Bilateral frontal, parietal lobes lesions	3.1	UWS	4 (0\0\1\1\0\2)
33	Female	69	CI	Right frontal, parietal lobes, basal ganglia lesions	6.9	UWS	5 (0\0\2\1\0\2)
34	Male	50	CI	Bilateral cerebral hemispheres extensive lesions, brainstem lesions	8.3	UWS	5 (0\0\2\1\0\2)
35	Male	50	TBI	Left cerebral hemisphere extensive lesions, brainstem lesions	10.0	UWS	6 (1\0\2\2\0\1)
36	Male	63	CH	Right frontal, parietal lobes lesions	45.0	MCS-	10 (1\1\5\1\0\2)
37	Male	53	CH	Bilateral parietal lobes lesions	15.7	MCS-	9 (1\3\1\2\0\2)
38	Female	67	CH	Bilateral frontal lobe, basal ganglia, right parietal lobe lesions	4.4	UWS	8 (1\1\2\2\0\2)
39	Male	55	CH	Left frontal, temporal lobes lesions	1.0	UWS	6 (0\1\2\1\0\2)
40	Male	47	TBI	Right cerebral hemisphere extensive lesions, bilateral cerebellar hemispheres, brainstem lesions	9.7	UWS	4 (0\0\1\1\0\2)
41	Male	51	CH	Bilateral frontal, occipital lobes, basal ganglia lesions	2.9	UWS	5 (0\0\2\1\0\2)
42	Male	28	HIE	Bilateral cerebral cortex lesions	5.1	UWS	5 (1\0\1\2\0\1)
43	Female	56	CI	Right cerebellar hemisphere extensive lesions, brainstem lesions	2.6	UWS	6 (0\1\2\1\0\2)
44	Male	23	TBI	Left cerebral cortex extensive lesions, right frontal lobe lesions	12.4	UWS	6 (0\1\2\1\0\2)
45	Female	70	HIE	Bilateral frontal, parietal lobes, basal ganglia, thalamus lesions	12.3	MCS-	9 (1\1\3\2\0\2)
46	Male	73	HIE	Bilateral cerebral hemispheres extensive lesions	4.3	UWS	4 (0\0\1\2\0\1)

Table 2 (continued)

Number	Gender	Age	Etiology	MRI/CT findings	Time since injury (months)	Diagnosis	CRS-R scores ^a
47	Female	57	CI	Right cerebral cortex lesions	3.6	UWS	6 (1\1\1\1\0\2)
48	Female	72	CH	Right parietal lobe, basal ganglia lesions	2.1	UWS	5 (0\1\1\1\0\2)
49	Female	58	CH	Left basal ganglia lesions	6.0	UWS	7 (1\1\2\1\0\2)
50	Female	69	CH	Right basal ganglia lesions	1.2	UWS	7 (1\1\2\1\0\2)
51	Male	73	CI	Bilateral frontal, parietal lobes, basal ganglia lesions	9.2	UWS	5 (0\0\2\1\0\2)
52	Female	59	CI	Left cerebral hemisphere extensive lesions	4.4	UWS	8 (1\1\2\2\0\2)

^aThe scores in the CRS-R scale are listed by subscale: respectively auditory, visual, motor, oromotor/verbal, communication, and arousal
CH, cerebral hemorrhage; CI, cerebral infarction; HIE, hypoxic ischemic encephalopathy; TBI, traumatic brain Injury; VS, vegetative state; UWS, unresponsive wakefulness syndrome; MCS, minimally conscious state; CRS-R, Coma Recovery Scale-Revised

covariate correction. The MCS group demonstrated higher FC in the following regions: PFC-L to PFC-R, SMC-L to SMC-R, and Wernicke-R to SMC-L (after FDR correction, Fig. 3A, B, Additional file 1: Fig. S3A). Channel-based analysis identified 24 significant connections between the MCS and VS/UWS groups before adjustment (after FDR correction, Additional file 1: Fig. S3B). These Ch-to-Ch differences were predominantly located within or between the following brain regions: within right PFC, left and right PFC, right PFC and left PMC, left PFC and right PMC, right PFC and left Wernicke's area, right PMC and left SMC, right PMC and left Wernicke's area, as well as right PMC and left VC. After adjusting for covariates, five significant connections remained, specifically between Ch3 to Ch11, Ch4 to Ch26, Ch4 to Ch29, Ch9 to Ch21, and Ch20 to Ch50. These reductions were primarily located in the left or right PFC and left PMC (after FDR correction, Fig. 3C, D). These five connections were subsequently selected for further analysis.

The correlation analysis between CRS-R scores and whole-brain features is presented in Fig. 4. FC between Ch3 and Ch11, Ch4 and Ch26, Ch4 and Ch29, and Ch20 and Ch50 exhibited significant correlations with CRS-R total scores (Fig. 4A–D). Regarding the CRS-R subscales, FC between Ch20 and Ch50 demonstrated a strong correlation with the visual subscale (Fig. 4E), while FC between Ch3 and Ch11, Ch4 and Ch26, Ch4 and Ch29 showed significant correlations with the motor subscale (Fig. 4F, G, H). Additionally, a significant correlation was found between the FC of left SMC to right SMC and the verbal subscale (Fig. 4I). No significant correlation was observed between whole-brain z-values or the following FC measures and the CRS-R total score or any of its subscales: FC between PFC-L and PFC-R, FC between Wernicke-R and SMC-L, and FC between Ch9 and Ch21.

The ROC curves and AUC values of whole-brain features for distinguishing MCS from VS/UWS are shown in Fig. 5, which include the average z-value (Fig. 5A), significant FC based on ROI to ROI (Fig. 5B), and significant FC based on Ch to Ch (Fig. 5C). Overall, the whole-brain

features effectively differentiated MCS from VS/UWS, with FC between Ch4-Ch29 and Ch9-Ch21 demonstrating good sensitivity and specificity, with AUC values of 0.818 and 0.817, respectively (Table 3). Moreover, FC between SMC-L and SMC-R, as well as the FC between Ch4 and Ch29, exhibited the highest performance accuracy among the ROI-ROI and Ch-Ch features, with accuracy rates of 69.23% and 76.92%, respectively (Table 3).

Seed-based network analysis

Connectivity was evaluated by calculating the strength of significant correlations between each seed region and the rest of the brain. Resting-state FC for the sensorimotor, auditory, frontoparietal, default mode and visual networks was observed in healthy participants, showing strong spatial consistency with prior fMRI studies [44, 45] (Additional file 1: Fig. S4). A detailed description of the brain regions linked to each network is provided in Table 1. All five networks demonstrated reduced connectivity in patients with MCS or VS/UWS compared to healthy controls. Furthermore, FC was significantly lower in patients with VS/UWS compared to those with MCS, particularly in the auditory, frontoparietal, and default mode networks (Table 1, Additional file 1: Fig. S4).

Regions exhibiting higher FC in MCS patients compared to VS/UWS patients within each network are presented in Additional File 1: Fig. S5. Individual connectivity values were extracted from each patient using binary masks that identified areas with higher connectivity in MCS compared to VS/UWS. These values were then averaged and used as features for the network. CRS-R total scores were found to correlate with the features from five networks (Fig. 6). Regarding the correlation between intrinsic connectivity networks and CRS-R subscales, significant associations were observed between features of the sensorimotor, auditory, and visual networks and the CRS-R visual subscale (Additional file 1: Fig. S6A, B, C). Additionally, features of the sensorimotor and visual networks also demonstrated significant

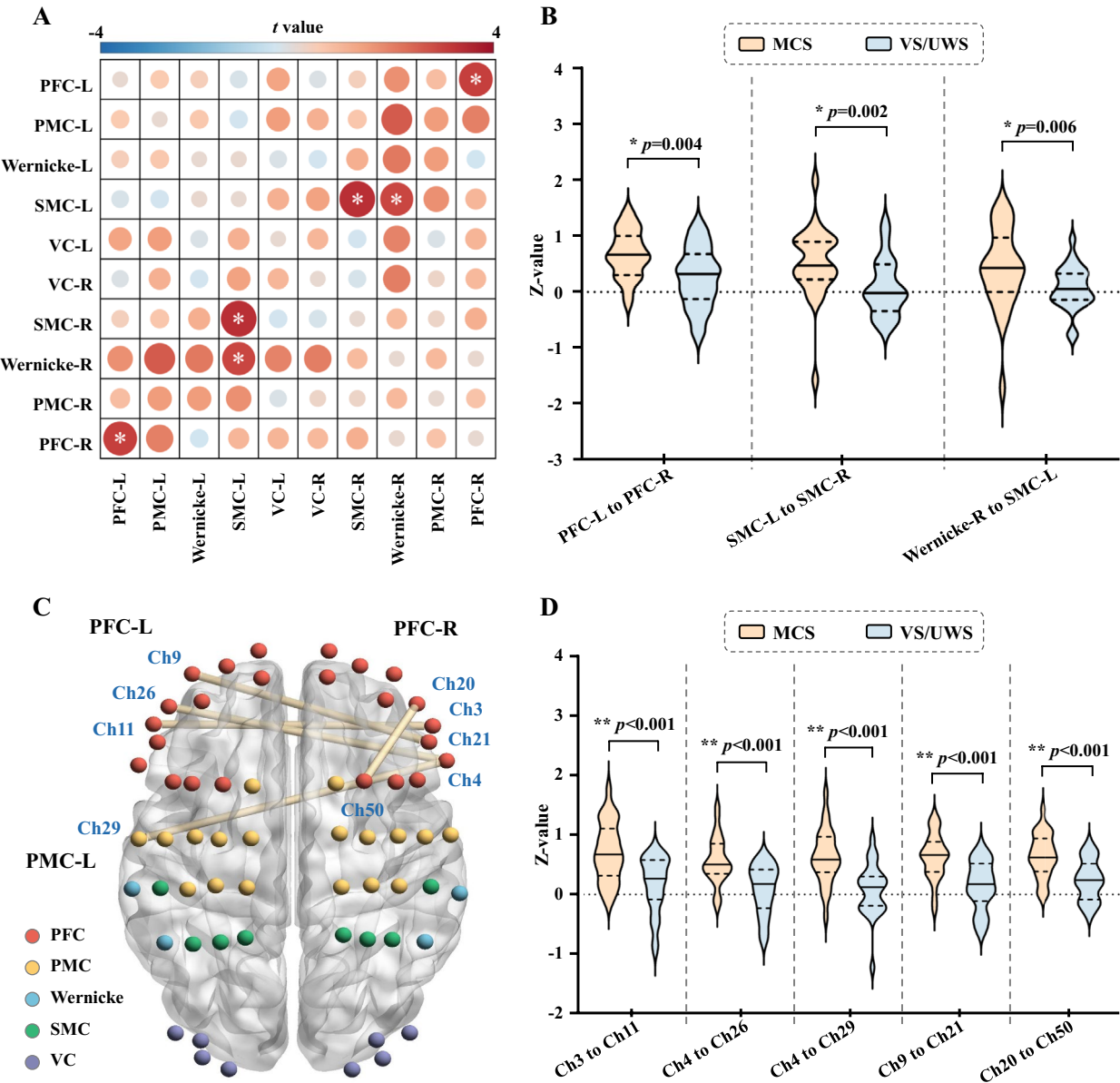


Fig. 3 Whole-brain functional connectivity analysis between patients with MCS and VS/UWS after adjustment. **A** T-value matrix of FC ROI-based FC, with * indicating significant FC between the two groups after adjusting for gender, age, etiology and time since injury ($p < 0.05$ after FDR correction); **B** Volin plots of significant ROI-ROI FC between two groups; **C** Significant Ch-based FC between two groups after adjusting for gender, age, etiology and time since injury; **D** Volin plots of significant Ch-Ch FC between two groups. Abbreviation: VS, vegetative state; UWS, unresponsive wakefulness syndrome; MCS, minimally conscious state; FC, functional connectivity; ROI, region of interest; PFC, prefrontal cortex; PMC, pre-motor cortex; SMC, sensorimotor cortex; VC, visual cortex; L, left; R, right; Ch, channel

correlations with the motor subscale (Additional file 1: Fig. S6D, E).

The ROC curves and AUC values of network features for distinguishing MCS from VS/UWS are presented in Fig. 5D and Table 3. The AUC values for the auditory, default, frontoparietal, sensorimotor, and visual networks were 0.803, 0.802, 0.800, 0.771, and 0.754, respectively, indicating strong sensitivity and specificity in differentiating MCS from VS/UWS. All networks demonstrated satisfactory accuracy in distinguishing MCS from VS/

UWS, with the SVM classification results aligning with the AUC findings. The auditory network performed the best, achieving an accuracy of 73.08%, followed by the default mode, frontoparietal, sensorimotor, and visual networks, with accuracy rates of 71.15%, 71.15%, 61.54%, and 57.69%, respectively (Table 3).

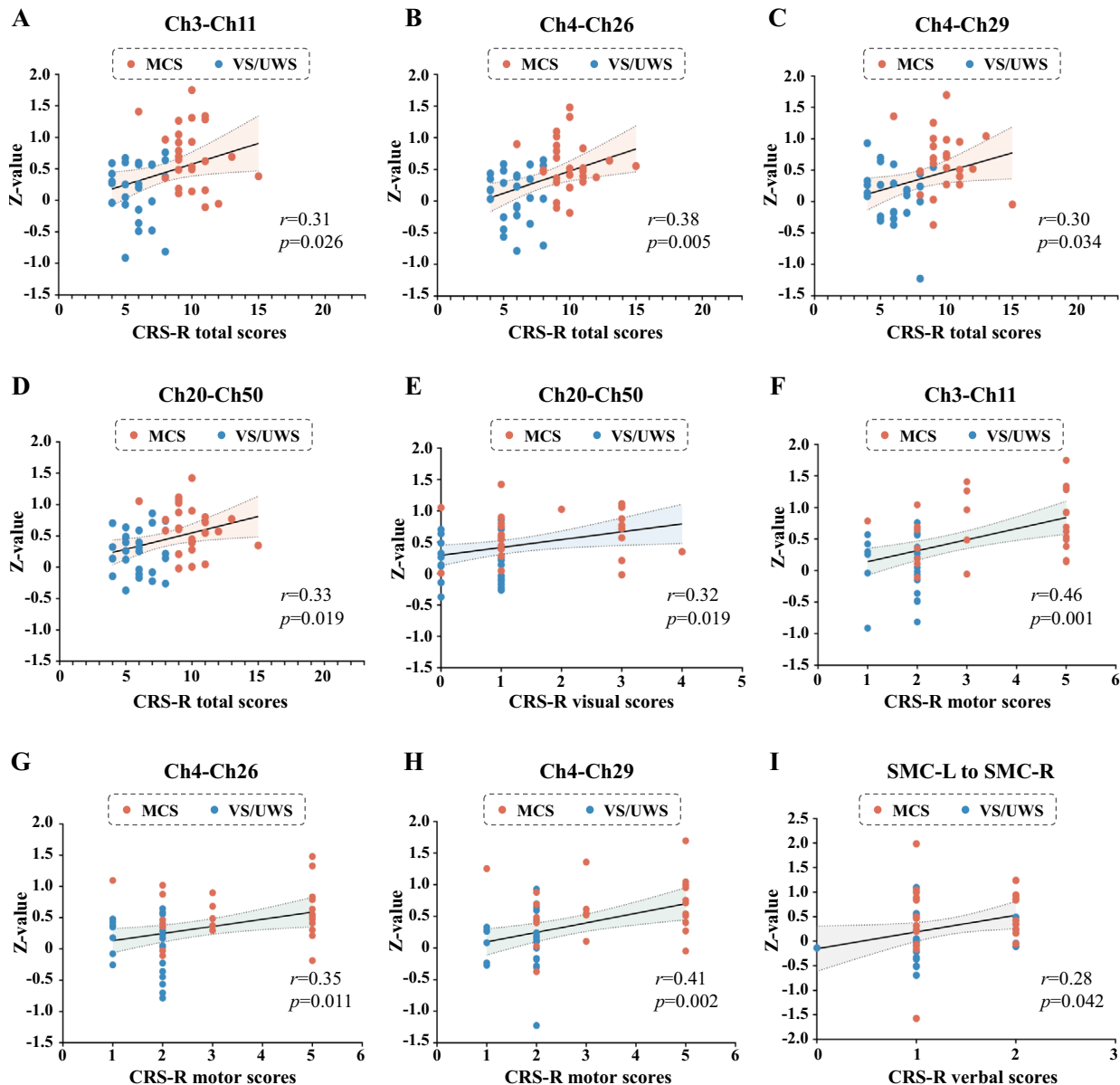


Fig. 4 Correlation analysis between CRS-R scores and whole-brain features in patients with DOC. **A** Correlation between CRS-R total scores and FC of channels 3 and 11; **B** Correlation between CRS-R total scores and FC of channels 4 and 26; **C** Correlation between CRS-R total scores and FC of channels 4 and 29; **D** Correlation between CRS-R total scores and FC of channels 20 and 50; **E** Correlation between CRS-R visual scores and FC of channels 20 and 50; **F** Correlation between CRS-R motor scores and FC of channels 3 and 11; **G** Correlation between CRS-R motor scores and FC of channels 4 and 26; **H** Correlation between CRS-R motor scores and FC of channels 4 and 29; **I** Correlation between CRS-R verbal scores and FC between left and right SMC. Abbreviation: VS, vegetative state; UWS, unresponsive wakefulness syndrome; MCS, minimally conscious state; FC, functional connectivity; SMC, sensorimotor cortex; L, left; R, right; Ch, channel; CRS-R, Coma Recovery Scale-Revised

Discussion

This study, using resting-state fNIRS with the largest sample size to date, examined whole-brain and seed-based network differences between DOC patients and healthy controls, as well as between MCS and VS/UWS patients. The key findings were: (1) DOC patients exhibited severe impairments in both whole-brain and network-based FC compared to healthy controls; (2) VS/UWS patients had significantly reduced FC compared to

MCS patients, including in the prefrontal cortex, sensorimotor regions, Wernicke's area, and within auditory, frontoparietal, and default mode networks, with some connectivity differences correlating with CRS-R scores; (3) Whole-brain and network connectivity from resting-state fNIRS effectively distinguished MCS from VS/UWS patients.

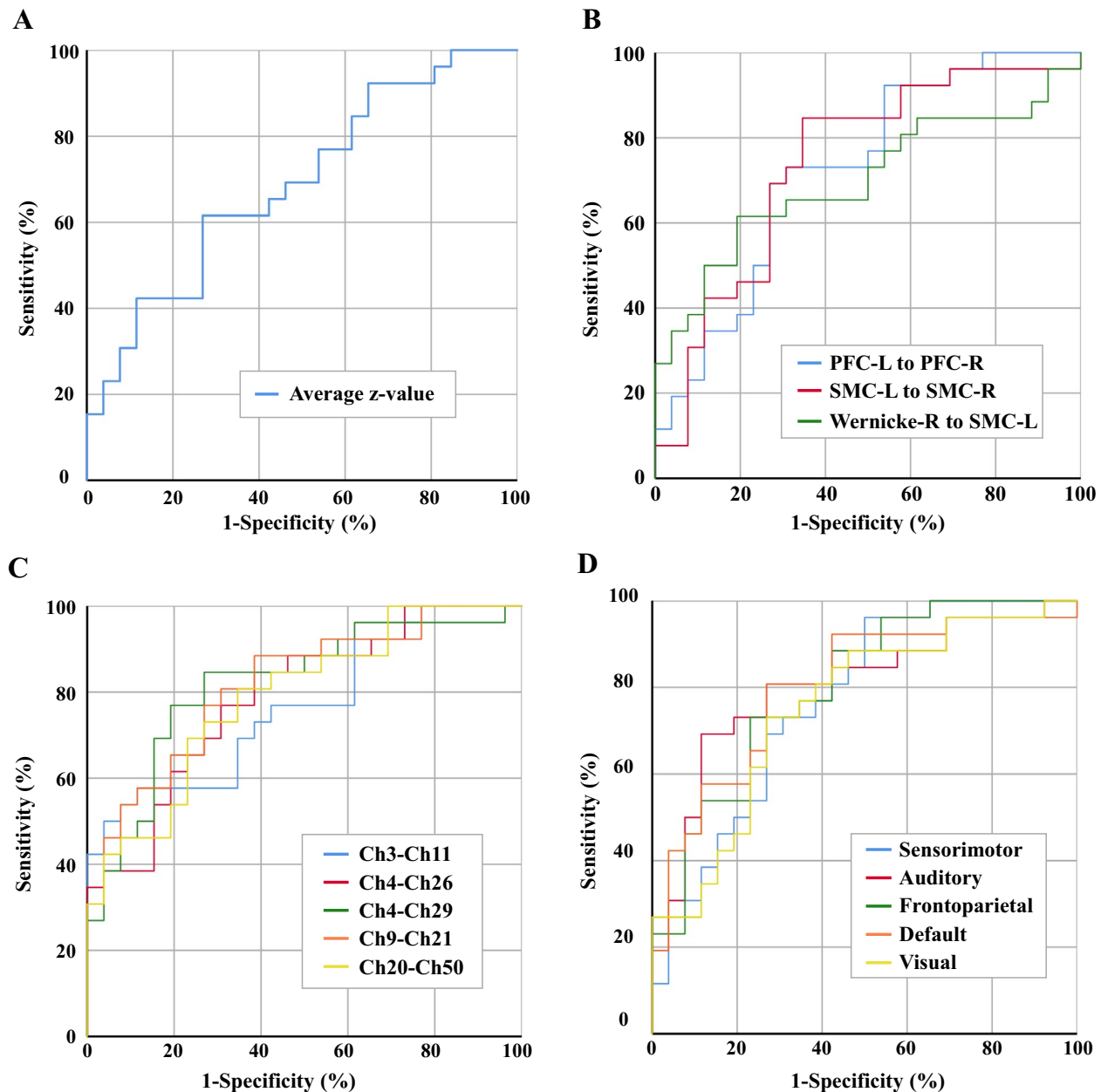


Fig. 5 ROC curves and classification accuracy for distinguishing MCS from VS/UWS patients. **A** ROC curve illustrating the relationship between average z-values and the diagnosis of MCS or VS/UWS; **B** ROC curves depicting FC based on ROIs in relation and their association with MCS or VS/UWS diagnosis; **C** ROC curves showing FC based on specific channels and their association with MCS or VS/UWS diagnosis; **D** ROC curves representing FC based on networks and their correlation with MCS or VS/UWS diagnosis. Abbreviation: ROC, Receiver Operating Characteristic; MCS, minimally conscious state; VS, vegetative state; UWS, unresponsive wakefulness syndrome; ROI, region of interest; Ch, channel; PFC, prefrontal cortex; SMC, sensorimotor cortex; L, left; R, right

Widespread disruption in functional connectivity for patients with DOC

The significantly reduced whole-brain and network FC in DOC patients suggested that abnormal brain connectivity might be a core feature of consciousness disorders, reflecting a loss of network integration.

Previous studies demonstrated that DOC patients exhibited distinct dynamic interactions between multiple

cortical and subcortical networks, depending on their condition severity [35, 46]. EEG studies reported significantly reduced FC, particularly in high-frequency bands (such as alpha and beta), closely associated with impaired consciousness levels [47, 48]. fMRI research further demonstrated widespread declines in networks like default mode network (DMN), frontoparietal, and sensorimotor network [11], with disruptions linked to consciousness

Table 3 Classification performance of resting-state features in differentiating MCS from VS/UWS

Features	AUC	Feature classification ^a		
		True positives (MCS)	True negatives (VS/UWS)	Performance accuracy (%)
Average z-value	0.689	23	8	59.62
ROI-ROI				
SMC-L to SMC-R	0.738	18	18	69.23
PFC-L to PFC-R	0.725	15	19	65.38
Wernicke-R to SMC-L	0.698	23	9	61.54
Ch-Ch				
Ch4-Ch29	0.818	20	20	76.92
Ch9-Ch21	0.817	19	18	71.15
Ch4-Ch26	0.788	17	19	69.23
Ch20-Ch50	0.788	15	19	65.38
Ch3-Ch11	0.766	17	16	63.46
Network				
Auditory	0.803	18	20	73.08
Default	0.802	18	19	71.15
Frontoparietal	0.800	14	23	71.15
Sensorimotor	0.771	13	19	61.54
Visual	0.754	9	21	57.69

^aA single-feature linear support vector machine classifier with leave-one-out cross-validation was used to evaluate the classification performance of the features in distinguishing MCS from VS/UWS patients

MCS, minimally conscious state; VS, vegetative state; UWS, unresponsive wakefulness syndrome; ROI, region of interest; Ch, channel; AUC, area under the curve; PFC, prefrontal cortex; SMC, sensorimotor cortex; L, left; R, right

levels. A meta-analysis also reported reduced activity in multiple brain regions in DOC patients compared to healthy controls, including the left cingulate gyrus, bilateral mediodorsal thalamus, and others [49]. Our findings aligned with fMRI [11, 50] and EEG studies [47, 48, 51], demonstrating widespread FC impairment in DOC patients, further validating the use of fNIRS for detecting brain activity and capturing common resting-state networks in both healthy individuals and DOC patients.

Several fNIRS-based studies reported abnormal FC in the prefrontal regions of DOC patients. For instance, Chen et al. [20] found widespread disruptions in frontal FC, particularly in the frontopolar and right dorsolateral prefrontal cortex in MCS patients. Liu et al. [16] further demonstrated that MCS and VS/UWS patients showed distinct topological structures and differences in short- and long-distance connectivity within the prefrontal cortex. Luo et al. [52] reported significant decreases in intra- and inter-network FC across multiple networks in DOC patients, such as the sensorimotor network, dorsal attention network, and default mode network. Our study extended these findings by showing widespread impairments in whole-brain FC, not just in the prefrontal cortex. With a larger sample size, it further validated the notion that DOC patients experience dysfunction in key

networks crucial for consciousness. This emphasized the importance of considering global brain connectivity in DOC assessment.

Difference in resting-state connectivity between MCS and VS/UWS patients

This study showed that VS/UWS patients had significantly reduced whole-brain and resting-state network FC compared to MCS patients, particularly in the DMN, frontoparietal network, and auditory-language regions. These results suggested that MCS patients retained some FC in higher-order networks, while VS/UWS patients exhibited a near-total loss of network integration [11].

The DMN is highly active during rest and supports introspection and self-referential processing. It is the most studied network in DOC imaging research [53]. A meta-analysis [49] using fMRI found that DMN connectivity was closely linked to consciousness levels, with VS/UWS patients showing nearly complete DMN integration loss, while MCS patients retained partial connectivity. The frontoparietal network, which integrated the frontal and parietal lobes to maintain conscious states, is also crucial in consciousness processing [54]. Neuroimaging studies have linked activation of the prefrontal and parietal cortices to consciousness states [55, 56]. Reduced FC in language and auditory areas might reflect a decline in language perception and conscious processing [55]. fMRI-based studies [57] indicated MCS patients might still respond to auditory stimuli, while VS/UWS patients showed minimal or no activity. Liang et al. [58] found reduced FC between the frontal and occipital lobes in VS/UWS patients compared to MCS and EMCS patients, while this study observed no occipital FC changes. By including multiple ROIs, this study captured broader brain network connectivity, highlighting FC differences in higher-order cognitive networks, rather than focusing on occipital FC changes. According to the Global Neuronal Workspace Theory (GNWT), consciousness arises from the extensive integration and real-time interaction of information across multiple brain regions [59]. He et al. [60] found that MCS patients exhibited higher global efficiency and shorter path lengths than UWS patients using resting-state fNIRS, which provided further support for GNWT. MCS patients might retain some connectivity in key consciousness-related areas, allowing for limited conscious processing. In contrast, VS/UWS patients showed widespread FC disruption, leading to a significant decline or loss of consciousness.

This study indicated that poorer consciousness levels, particularly impaired visual and motor functions, were associated with weaker connectivity within these regions and networks. Previous studies also reported correlations between brain network FC and behavioral assessments. Demertzi et al. [11] highlighted the link between

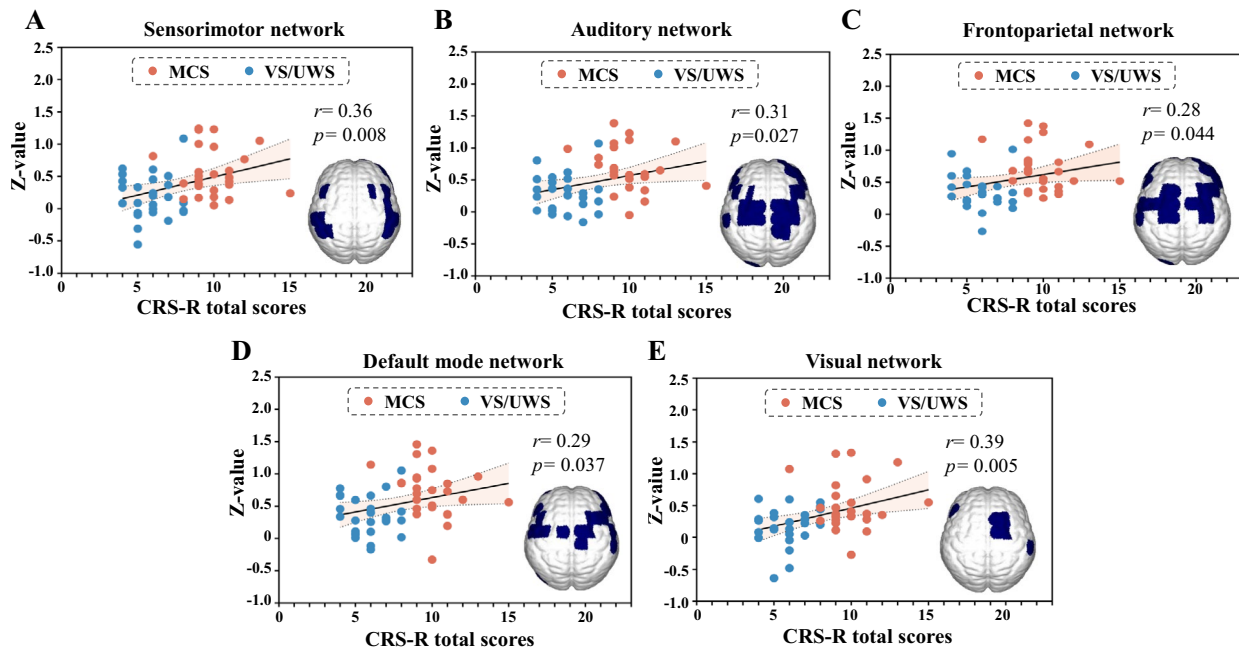


Fig. 6 Correlation between CRS-R total scores and seed-based network connectivity analysis in patients with MCS and VS/UWS. **A** Correlation between CRS-R total scores and FC of sensorimotor network; **B** Correlation between CRS-R total scores and FC of auditory network; **C** Correlation between CRS-R total scores and FC of frontoparietal network; **D** Correlation between CRS-R total scores and FC of default mode network; **E** Correlation between CRS-R total scores and FC of visual network. The functional connectivity of the difference maps (areas in blue) between the MCS and VS/UWS groups was used for the correlation analysis. Abbreviation: MCS, minimally conscious state; VS, vegetative state; UWS, unresponsive wakefulness syndrome; CRS-R, Coma Recovery Scale-Revised

resting-state networks and CRS-R scores in DOC patients. A resting-state fNIRS study [20] found that the nodal topological properties of the right DLPFC were positively correlated with auditory subscale scores. We did not find a significant correlation with the auditory subscale, likely because most MCS patients were unable to follow commands, with only one MCS+ patient scoring 3 points.

To ensure stable placement of the fNIRS optodes and signal quality, all patients were positioned with the head of the bed elevated by 30° and supported with a neck pillow [13]. Current research suggested that verticalization might improve consciousness or lead to orthostatic hypotension and circulatory changes, which could influence brain function assessments [61, 62]. Since all participants underwent the same elevation during the fNIRS evaluation, posture was unlikely to have a significant effect on the results. Etiology and time since injury were included as covariates to identify more robust connections, but their impact remained unclear. Chen et al. [39] found better prognoses in traumatic brain injury (TBI) patients, while hypoxia-induced DOC had poorer outcomes. An EEG study [40] showed that hypoxic patients had lower firing rates, while stroke and brain injury patients had higher measures. DOC duration was also positively correlated with EEG measures. However, the use of fNIRS in

assessing consciousness under varying clinical conditions needs further validation.

Discriminative power of fNIRS for VS/UWS and MCS

The results showed that, among the whole-brain features, Ch4-Ch29 demonstrated the highest accuracy. This long-distance connection across hemispheres might reflect the critical role of coupling in consciousness processing [63]. Among resting-state brain network metrics, the auditory network showed the highest accuracy, suggesting that impaired auditory perception and sound processing might be characteristics of low-level consciousness disorders [2].

Several studies have explored various techniques to differentiate MCS and VS/UWS patients. Fluorodeoxyglucose PET (FDG-PET) achieved 85% consistency and 74% prediction accuracy [64]. A meta-analysis [65] reported that fMRI showed a combined sensitivity of 0.71, a specificity of 0.71, and an AUC of 0.76. Machine learning has improved the precision of neuroimaging and electrophysiological techniques in assessing DOC patients' consciousness levels [66]. Sheng et al. [67] used nonlinear dynamic analysis of EEG based on auditory stimuli, achieving accuracy from 61.11 to 83.33%. The potential of fNIRS in differentiating levels of consciousness has been highlighted. Kazazian et al. [13], using resting-state fNIRS, found that the sensitivity for a single

healthy subject in the resting networks ranged from 65.2 to 73.9%. Othman et al. [68] demonstrated that the neurovascular coupling in the frontal region based on fNIRS and EEG signals could distinguish VS/UWS patients from awake brain injury patients with an accuracy of over 90%.

Compared to previous studies, our findings suggested that whole-brain and brain network features based on resting-state fNIRS exhibited acceptable accuracy. Previous fMRI studies [50, 69] achieved accuracies ranging from 80 to 88% with an AUC of 0.83. In contrast, our fNIRS-based classification model reached a maximum accuracy of 76.92% and an AUC of 0.818, slightly lower than fMRI. This difference may be due to fNIRS detecting only cortical activity, while subcortical structures critical in consciousness disorders [70], were not included. Nonetheless, as a portable and low-cost technique, fNIRS still offered unique advantages in clinical settings, particularly when fMRI was not feasible, fNIRS might serve as a valuable supplementary diagnostic tool.

Limitations

This study had several limitations. First, brain ischemia or hematomas in some patients could affect light scattering and absorption, and brain reorganization in DOC patients might influence probe placement due to shifts in brain regions. Although we recorded the damaged areas from CT/MRI, ischemic and hematoma lesions were difficult to quantify. Second, the spatial resolution of fNIRS was relatively low, which might limit its ability to detect FC changes in deeper brain regions. Third, fNIRS changes might vary with etiology and disease duration. Future studies should explore how these factors influence fNIRS features and examine differences between MCS and UWS patients with varying clinical factors. Forth, Mandrekar [71] defines AUC values of 0.7–0.8 as acceptable and 0.8–0.9 as excellent. In our dataset, AUC for whole-brain average FC and ROI-based FC (both < 0.75) might not reliably classify individual consciousness levels, highlighting clinical limitations. However, some CH-based and network FC measures showed AUC values > 0.8, indicating better discriminative power and highlighting the varying effectiveness of different fNIRS measures in assessing DOC patients. Finally, this study was based solely on resting-state data. Future research should incorporate task-based fNIRS to explore dynamic brain network characteristics, prognosis, and long-term outcomes. Combining fNIRS with other neuroimaging techniques and advanced machine learning could improve sensitivity and provide more comprehensive assessments.

Conclusion

Resting-state fNIRS could detect abnormal whole-brain and network connectivity in DOC patients. MCS patients showed preserved FC in higher-order networks, while VS/UWS patients exhibited significant network disruption. fNIRS effectively distinguished between MCS and VS/UWS, offering insights for clinical diagnostics and monitoring. Future research should explore fNIRS variations across DOC patients with different clinical factors and combine task-based and other neuroimaging techniques to assess its prognostic value.

Abbreviations

AUC	Area under the curve
Ch	Channel
CRS-R	Coma recovery scale-revised
CV	Coefficient of variation
DMN	Default mode network
DOC	Disorders of consciousness
EEG	Electroencephalography
FC	Functional connectivity
FDG-PET	Fluorodeoxyglucose positron emission tomography
FDR	False discovery rate
fMRI	Functional magnetic resonance imaging
fNIRS	Functional near-infrared spectroscopy
GNWT	Global neuronal workspace theory
HbO	Oxygenated hemoglobin
HbR	Deoxygenated hemoglobin
HbT	Total hemoglobin
HC	Healthy controls
MCS	Minimally conscious state
OD	Optical density
PFC	Prefrontal cortex
PMC	Pre-motor cortex
ROC	Receiver operating characteristic
ROI	Region of interest
SMC	Sensorimotor cortex
SVM	Support vector machine
TBI	Traumatic brain injury
UWS	Unresponsive wakefulness syndrome
VC	Visual cortex
VS	Vegetative state

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-025-07181-z>.

Additional file 1.

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Author contributions

SW conceptualized and designed the study, collected and analyzed the data, drafted the initial manuscript. BZ and ZY conceptualized and designed the study, collected and analyzed the data, reviewed and edited the manuscript. HC, MY, and YA were responsible for data analysis and for reviewing and editing the manuscript. XY, FZ, ZY, AC, and RW contributed to data collection and manuscript revision. YZ, HD, NH, XH, and LZ supervised and directed the experimental implementation.

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Data availability

The data analyzed in this study are not publicly available but can be obtained from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The research involving human participants was approved by the Ethics Committee of the Third Affiliated Hospital of Sun Yat-Sen University (Approval code: RG2025-039). All patients/participants provided written informed consent to take part in the study. Written consent was also obtained for the publication of any potentially identifiable images or data in this article.

Consent for publication

Not applicable.

Competing interests

The authors declare no potential conflicts of interest or personal relationships that could have influenced the work reported in this paper.

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